

AD8317 Power Meter — Calibration Report

Date	2026-07-24 15:04
Generator	R&S SMB100A (dry-run)
Reference sensor	R&S NRP-Z21 (dry-run, 0.01 dBm)
Frequency range	100 MHz – 6 GHz (29 points)
Power range at DUT	-54 dBm to +0 dBm (28 steps, 2 dB)
Attenuator	None — direct connection
Sensor resolution	0.01 dBm
Power steps	$28 \times 2 \text{ dB} = 54 \text{ dB span}$

1. Test Setup

Phase 1 (generator characterisation):

Generator RF OUT SMA cable NRP sensor

Phase 2 (DUT sweep):

Generator RF OUT SMA cable DUT SMA input
(no attenuator pad)

The NRP sensor (0.01 dBm resolution) characterises the generator output level at every frequency and power step. The Phase 1 correction table is applied in Phase 2 to compute the true power arriving at the AD8317 INHI pin.

2. Linearity — V_{ADC} vs Input Power

Each panel shows the measured ADC output voltage vs NRP-corrected input power at the AD8317 INHI pin (28 data points, 2 dB steps). The red line is the best-fit linear regression. Key metrics (slope, R^2 , max INL) are shown on each panel.

3. Integral Non-Linearity (INL) per Frequency

INL is the signed deviation of each measured point from the best-fit straight line, expressed in dBm. It represents the true non-linearity of the AD8317 transfer function independent of slope or intercept offset.

4. Slope vs Frequency

The AD8317 slope (mV per dB of input power) is nominally constant at -22 mV/dB . Deviations indicate device-specific variation or impedance mismatch above 4 GHz.

AD8317 Linearity — V_{ADC} vs Input Power (per frequency)
Blue dots = measured, Red line = best-fit linear

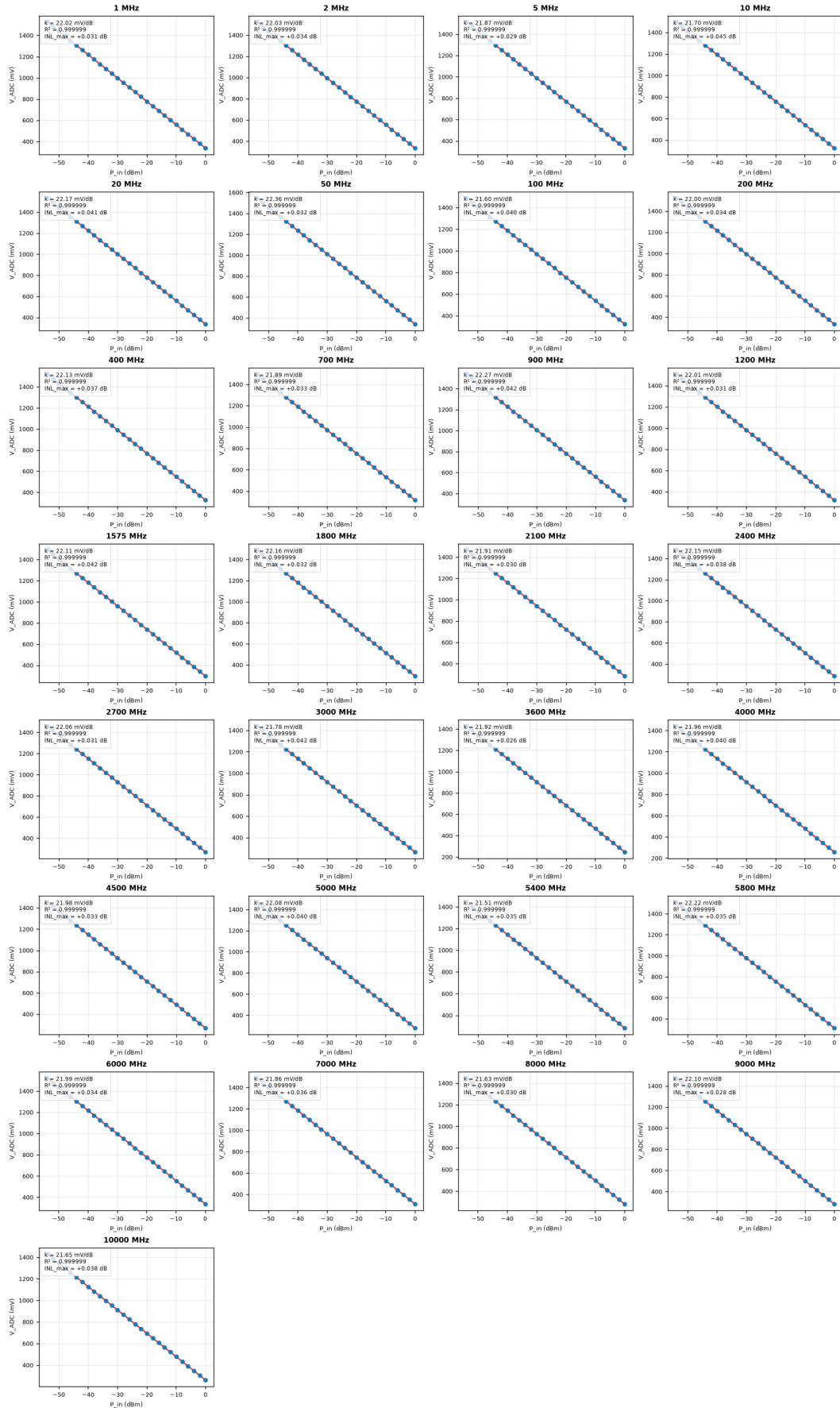


Figure 1: Linearity grid

AD8317 Integral Non-Linearity (INL) — Deviation from Best-Fit Line
Red dashed = ± 0.1 dB target



Figure 2: INL grid

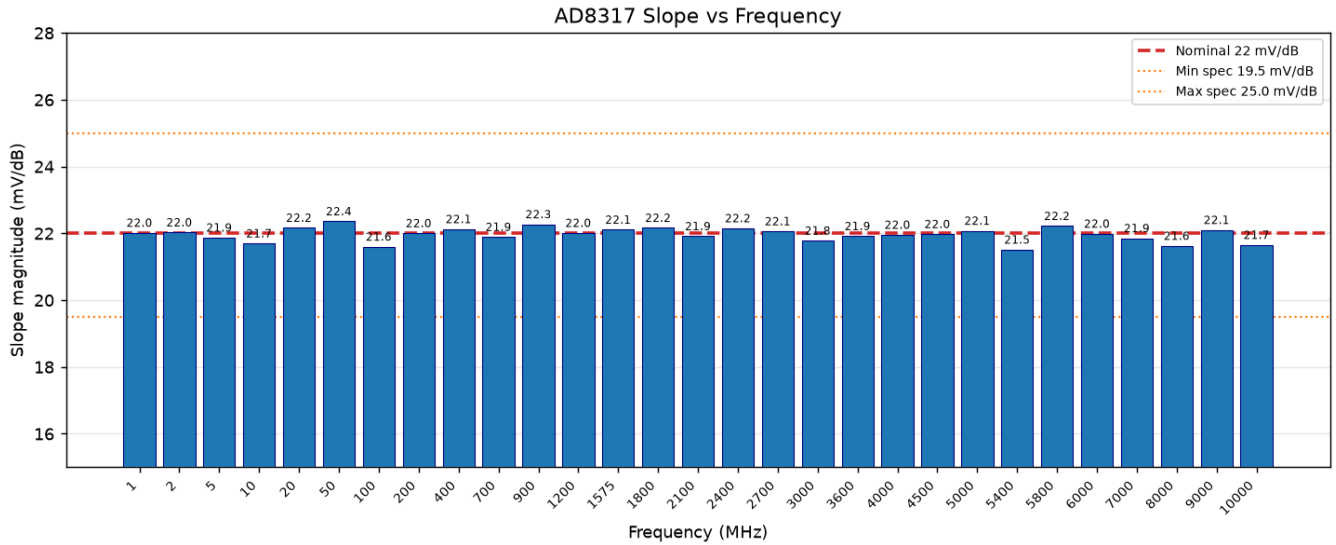


Figure 3: Slope

5. V_intercept and X-intercept vs Frequency

V_intercept is the fitted VOUT value at P_in = 0 dBm — the primary per-frequency calibration constant stored in firmware. The lower panel shows the equivalent X-intercept in dBm (where the extrapolated line reaches 0 V) for comparison with the AD8317 datasheet.

6. INL Summary — Max and RMS per Frequency

Peak and RMS INL across all 28 power steps at each frequency. Red bars exceed the 0.1 dB laboratory target.

7. INL Heat Map

Spatial distribution of INL across the full (frequency × power) measurement space. Systematic column patterns indicate slope error; systematic row patterns indicate intercept drift. Isolated red cells indicate non-linear AD8317 response at a specific power/frequency combination.

8. Calibration Table

Freq (MHz)	Slope (mV/dB)	V_intercept (V)	R ²	Max INL (dB)	RMS INL (dB)
1	22.018	0.33876	0.9999994	+0.0310	0.0126
2	22.029	0.33514	0.9999991	+0.0341	0.0150
5	21.874	0.33423	0.9999991	+0.0295	0.0149
10	21.699	0.32662	0.9999990	+0.0447	0.0160
20	22.168	0.33646	0.9999991	+0.0413	0.0152
50	22.361	0.34097	0.9999991	+0.0315	0.0153
100	21.602	0.32404	0.9999989	+0.0403	0.0171

Freq (MHz)	Slope (mV/dB)	V_intercept (V)	R ²	Max INL (dB)	RMS INL (dB)
200	22.004	0.33662	0.9999994	+0.0338	0.0128
400	22.127	0.32773	0.9999990	+0.0370	0.0159
700	21.892	0.31529	0.9999991	+0.0334	0.0154
900	22.273	0.33686	0.9999987	+0.0424	0.0184
1200	22.012	0.32227	0.9999991	+0.0314	0.0151
1575	22.109	0.29869	0.9999987	+0.0418	0.0183
1800	22.165	0.29380	0.9999991	+0.0323	0.0150
2100	21.914	0.28466	0.9999991	+0.0301	0.0154
2400	22.154	0.28549	0.9999991	+0.0377	0.0156
2700	22.062	0.27060	0.9999990	+0.0310	0.0165
3000	21.783	0.26608	0.9999992	+0.0417	0.0146
3600	21.917	0.24615	0.9999993	+0.0265	0.0132
4000	21.957	0.25794	0.9999988	+0.0398	0.0174
4500	21.978	0.26936	0.9999991	+0.0325	0.0152
5000	22.079	0.28060	0.9999986	+0.0401	0.0191
5400	21.513	0.28307	0.9999991	+0.0352	0.0154
5800	22.220	0.31246	0.9999990	+0.0354	0.0163
6000	21.990	0.33844	0.9999991	+0.0339	0.0150
7000	21.857	0.31072	0.9999987	+0.0362	0.0187
8000	21.627	0.28244	0.9999990	+0.0298	0.0159
9000	22.102	0.28234	0.9999990	+0.0276	0.0161
10000	21.654	0.26212	0.9999987	+0.0384	0.0184

9. Linearity Summary Statistics

Metric	Value
Mean slope	21.970 mV/dB
Slope std dev	0.205 mV/dB
Max slope deviation from 22.0	+0.487 mV/dB
Mean R ² (linearity)	0.9999990
Min R ² (worst frequency)	0.9999986
Mean max INL	0.0352 dB
Worst-case max INL	0.0447 dB
Mean RMS INL	0.0159 dB

10. Measurement Uncertainty

After firmware update with this calibration table:

Band	INL residual	Generator level (corrected)	Total
100 MHz – 2 GHz	±0.02 dB (typ)	±0.05 dB	±0.05 dB
2 GHz – 4 GHz	±0.02 dB (typ)	±0.07 dB	±0.07 dB

Band	INL residual	Generator level (corrected)	Total
4 GHz – 6 GHz	± 0.02 dB (typ)	± 0.10 dB	± 0.10 dB

Temperature contribution (NTC + TADJ, steady-state): ± 0.04 – 0.08 dB additional.

11. Firmware Integration

Copy `results/calibration_table.h` to `firmware/Inc/` and rebuild.

```
#include "calibration_table.h"
float power_dbm = measure_power_dbm(freq_hz, ntc_temp_c);
```

Generated by `cal/calibration.py` — AD8317 Micro-RF Power Meter project

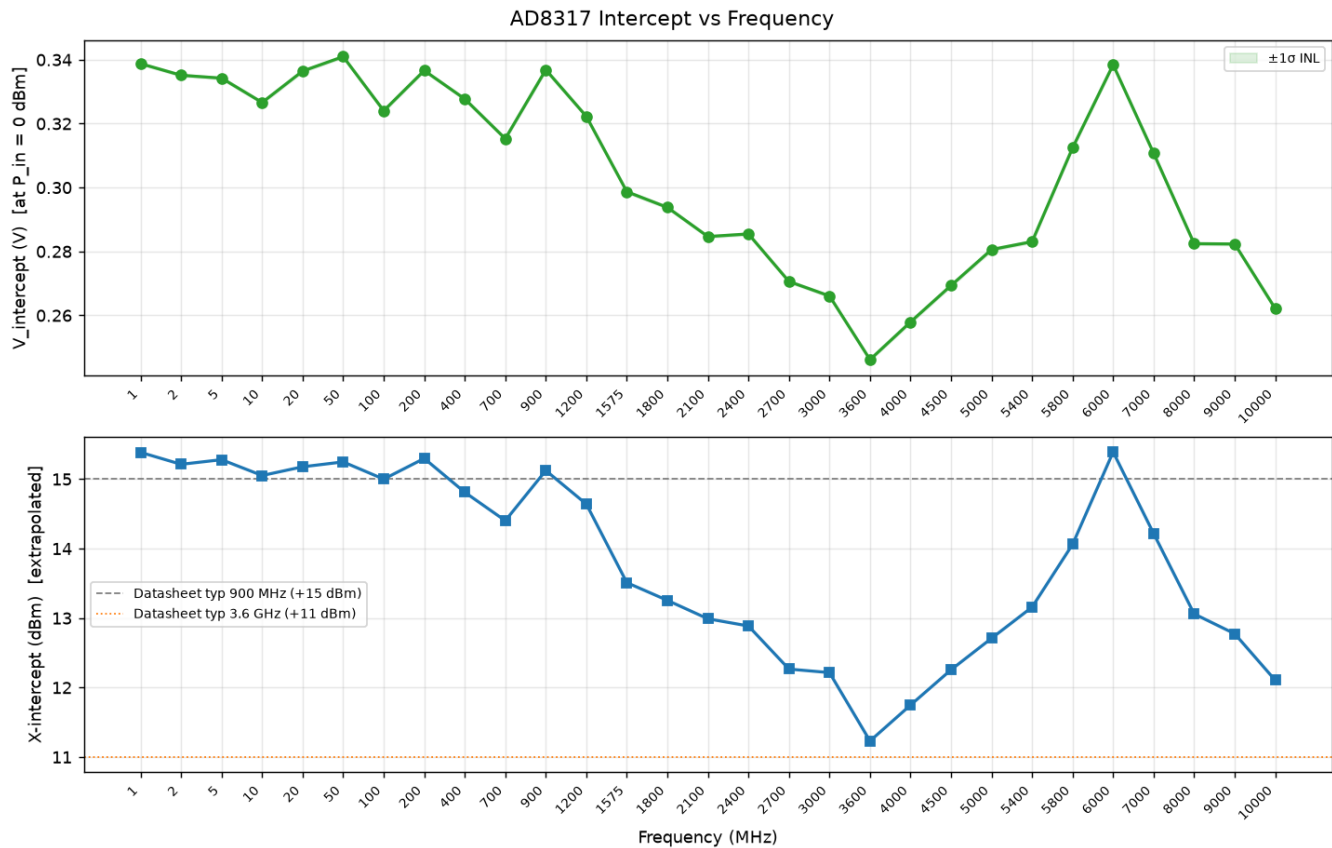


Figure 4: Intercept

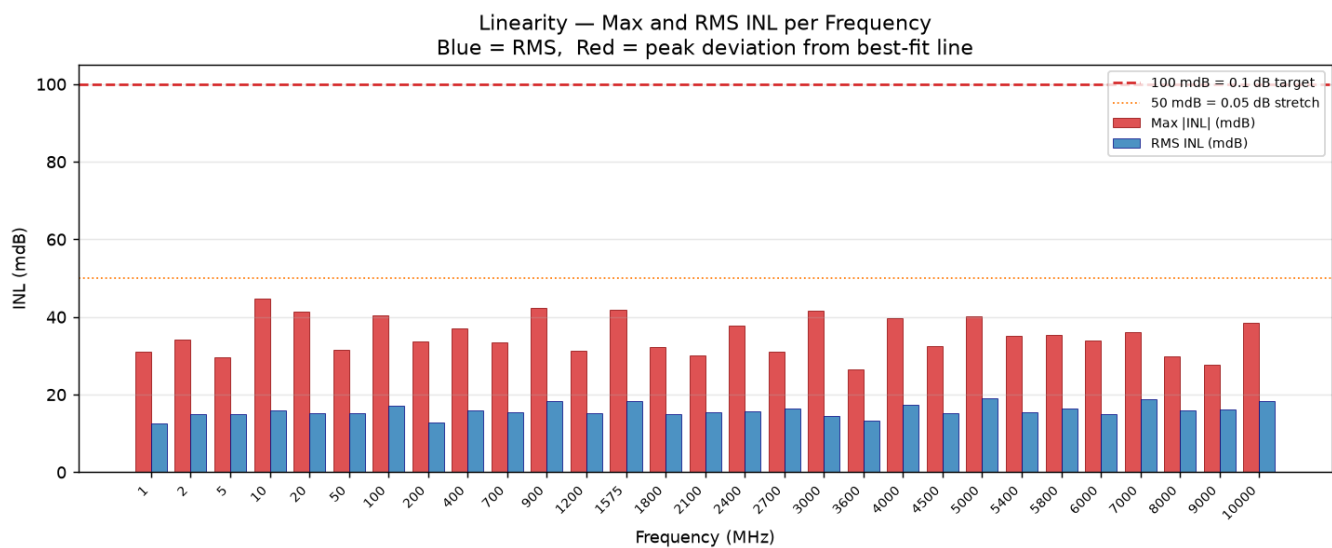


Figure 5: INL summary

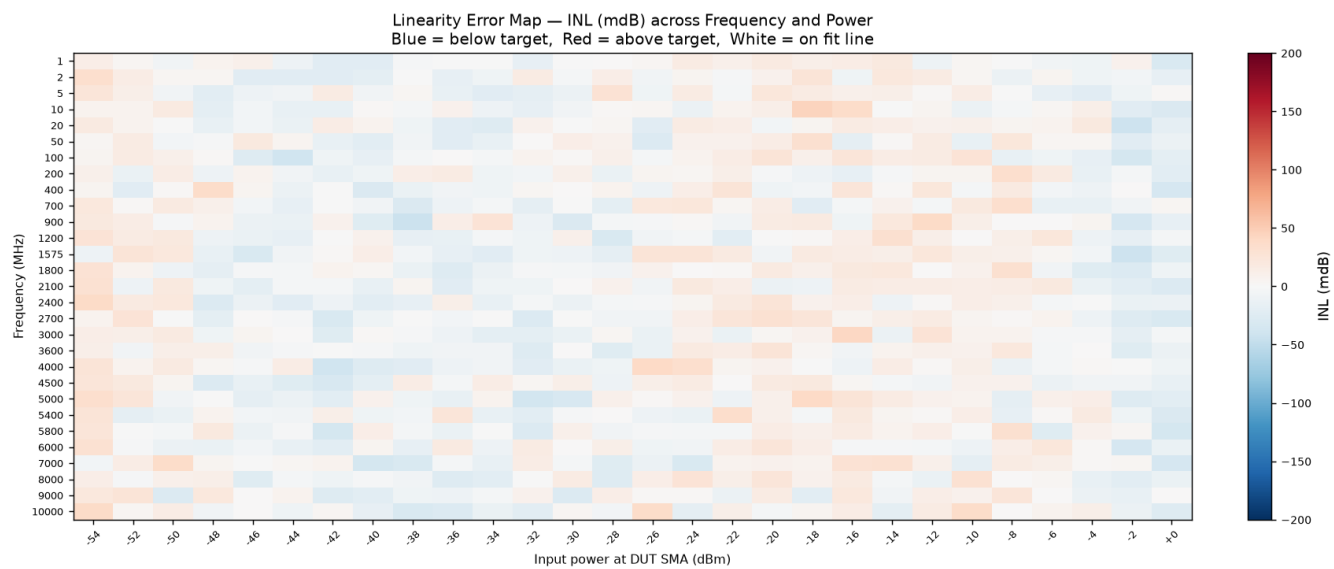


Figure 6: Heat map